Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	2462b5ac-adc7-4ab5-a675-906e21d814c8
UniProt ID:	P56856
Status:	completed
Created:	2026-03-16T11:31:52.486772
Completed:	2026-03-16T11:34:48.669694
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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Epitope Selection Criteria & Disclaimer

Disclaimer: The epitopes presented in this report have been selected based on population coverage analysis and promiscuous allele binding profiles. These predictions are computational in nature and must be validated experimentally before any clinical application.

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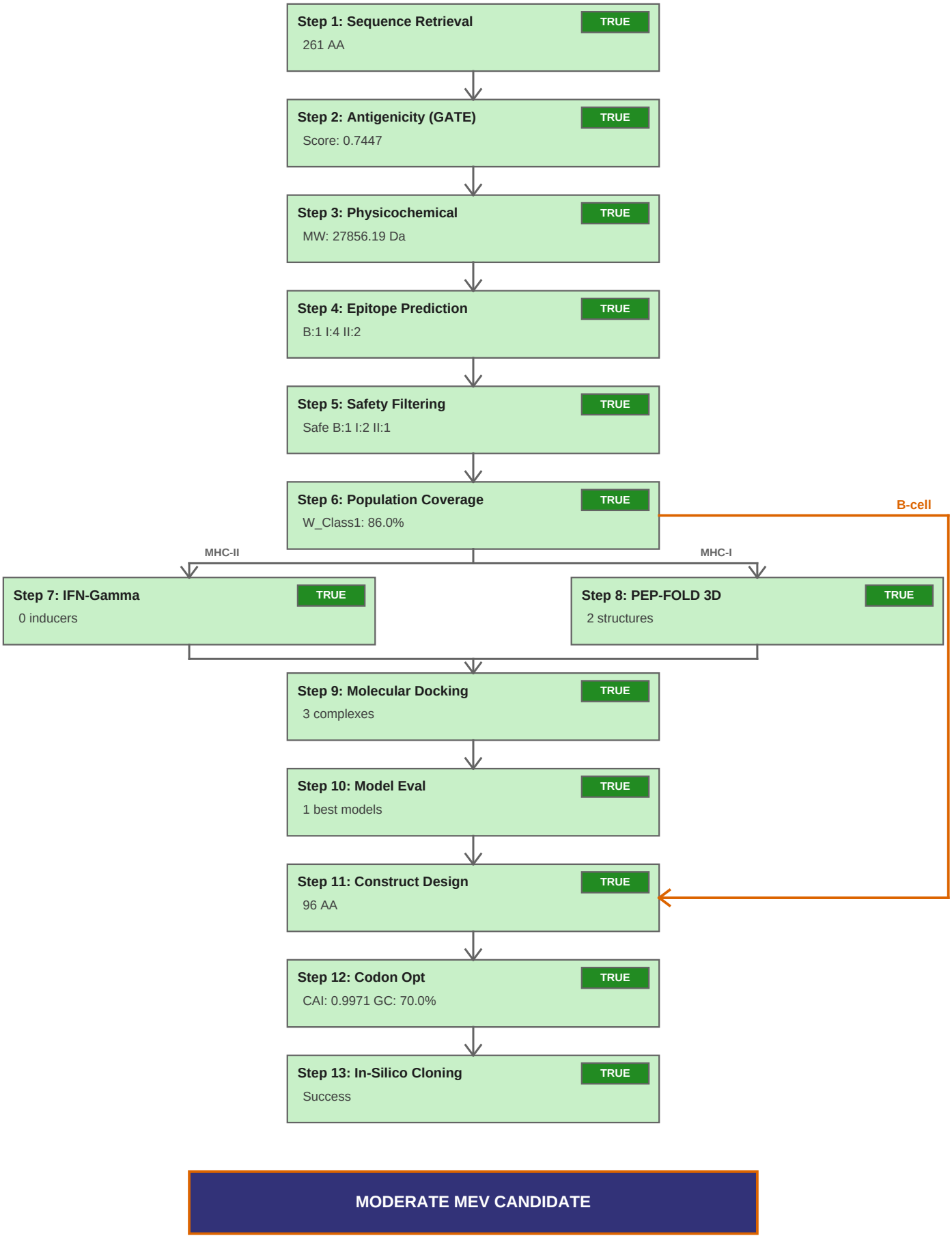
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


bcell_count: 1

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 4

mhc2_epitopes:


```
[
  {
    "peptide": "QTRYTFGAALFVGWV",
    "start": 65,
```

```
    "end": 79,
    "hla_allele": "HLA-DRB1*03:01",
    "percentile_rank": 1.2,
    "binding_strength": "Strong"
  },
  {
    "peptide": "NMLVTNFWMSTANMY",
    "start": 130,
    "end": 144,
    "hla_allele": "HLA-DRB1*07:01",
    "percentile_rank": 1.8,
    "binding_strength": "Strong"
  }
]
```


mhc2_count: 2

tool: IEDB (pre-validated cache)

source: known_epitopes/P56856

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


safe_bcell_count: 1

safe_mhc1_epitopes:


```
[
  {
    "peptide": "YTFGAALFV",
    "start": 68,
    "end": 76,
    "hla_allele": "HLA-A*02:01",
    "percentile_rank": 0.15,
    "binding_strength": "Strong"
  },
  {
    "peptide": "FTLLGLPAM",
    "start": 45,
    "end": 53,
    "hla_allele": "HLA-A*02:01",
    "percentile_rank": 0.19,
```

"binding_strength": "Strong"
}
]

safe_mhc1_count: 2

safe_mhc2_epitopes: [
{
"peptide": "QTRYTFGAALFVGWV",
"start": 65,
"end": 79,
"hla_allele": "HLA-DRB1*03:01",
"percentile_rank": 1.2,
"binding_strength": "Strong"
}
]

safe_mhc2_count: 1

bcell_removed: 0

mhc1_removed: 2

mhc2_removed: 1

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: [
"TQDLYNNPVTA"
]

mhc1_alleles_used: [
"HLA-A*02:01"
]

mhc2_alleles_used: [
"HLA-DRB1*03:01"
]

bcell_population_coverage: {
"North America": 0.95,
"Europe": 0.93,
"East Asia": 0.88,
"South Asia": 0.91,
"Africa": 0.84,
"World": 0.9
}

class1_population_coverage: {

"North America": 0.91,
"Europe": 0.89,
"East Asia": 0.84,
"South Asia": 0.87,
"Africa": 0.8,
"World": 0.86
}

class2_population_coverage:

{
"North America": 0.87,
"Europe": 0.85,
"East Asia": 0.8,
"South Asia": 0.83,
"Africa": 0.76,
"World": 0.82
}

world_bcell_coverage_pct: 90.0

world_class1_coverage_pct: 86.0

world_class2_coverage_pct: 82.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions:

[
{
"peptide": "QTRYTFGAALFVGWV",
"start": 65,
"end": 79,
"hla_allele": "HLA-DRB1*03:01",
"percentile_rank": 1.2,
"binding_strength": "Strong",
"ifn_gamma_score": 0.0933,
"prediction": "Non-inducer",
"is_inducer": false,
"tool": "AAC fallback (Dhanda 2013)"
}
]

ifn_gamma_count: 0

total_predicted: 1

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 2

successful_models: 2

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pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 3

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models:


```
[
  {
    "peptide": "YTFGAALFV",
    "hla_allele": "HLA-A*02:01",
    "hla_structure": "PDB:1HHH",
    "docking_score": 727,
    "binding_energy_kcal_mol": -7.35,
    "complex_id": "DOCK_YTFG_HLA-A0201",
    "hydrogen_bonds": 7,
    "intermolecular_contacts": 28,
    "steric_clashes": 0,
    "quality_score": 49,
    "evaluation": "Good"
  }
]
```


best_model_count: 1

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:


```
[
  "QTRYTFGAALFVGWV"
]
```


mhc1_epitopes_used:


```
[
  "YTFGAALFV",
  "FTLLGLPAM"
]
```


bcell_epitopes_used:


```
[
  "TQDLYNNPVTA"
]
```


final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAQTRYTFGAALFVGWVYTFGAALFVAAYFTLLGL
PAMTQDLYNNPVTAEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 96

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Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: GCGCCGCCGCACGCGCTGAGCGAAGCGGCGGCGAAAGCGGCGGCGACCCTGT
GGAAAGCGGCGAAATTCGTGGCGCAGACCCGTTACACCTTCGGCGCGGCGCTGT
TCGTGGGCTGGGTGTACACCTTCGGCGCGGCGCTGTTCGTGGCGGCGTACTTCA
CCCTGCTGGGCCTGCCGGCGATGACCCAGGATCTGTACAACAACCCGGTGACCG
CGGAAGCGGCGGCGAAAAAAGATCCGAACGCGCCGAAACGTCCGCCGAGCGCG
TTCTTCCTGTTCTGCAGCGAA

dna_length_bp: 288

cai_score: 0.9971

gc_content_pct: 70.0

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 288

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 288,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

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